

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

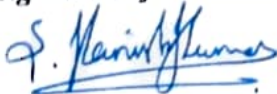
Annexure A

Analytical Problem Solving

Code of Conduct:

I Kanish Kumar.S (192511415) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



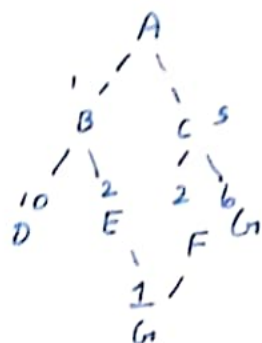


C01
ASSESSMENT TOOL - 1

Trace the working of Breadth-First search and Uniform Cost Search on a weighted graph with at least 7 nodes where edge costs are unequal. Show why BFS may not find the optimal path while UCS does, using your example graph.

Ans)

EXAMPLE WEIGHTED GRAPH:



EDGE COSTS:

- $A \rightarrow B = 1$
- $A \rightarrow C = 5$
- $B \rightarrow D = 10$
- $B \rightarrow E = 2$
- $C \rightarrow F = 2$
- $C \rightarrow G = 6$
- $E \rightarrow G = 1$

Start Node : A

Goal Node : G

BRENDTH FIRST SEARCH:

BFS explores nodes level by level without considering edge costs.

Traversal:

1. A
2. B, C
3. D, E, F, G (Goal Found)

Path found : $A \rightarrow C \rightarrow G$

Total cost : $5 + 6 = 11$

UNIFORM COST SEARCH (UCS):

UCS expands the node with the lowest cumulative path cost.

STEP	EXPANDED NODE	PATH COST
1	A	0
2	B	1
3	E	3
4	G	4 (Goal)

Optimal Path : $A \rightarrow B \rightarrow E \rightarrow G$

Total cost : $1 + 2 + 1 = 4$

CONCLUSION:

→ BFS finds the path with the fewest edges, not the minimum cost.

→ UCS always finds the least-cost path by considering cumulative edge costs.

For a chess-playing agent, characterize the task environment across all six PEAS/ environment dimensions. Justify each classification with reference to the rules of chess.

Ans)

CHESS - PLAYING AGENT : TASK ENVIRONMENT

- A chess playing agent operates in a fully observable environment because both players can see the entire chessboard and the positions of all pieces at all times.
- The environment is deterministic because every move produces a predictable result according to the fixed rules of chess. There is no element of chance or randomness.
- The environment is sequential because each move affects the future state of the game. A player's current decision influences all subsequent moves and the final outcome.
- The environment is static because the chessboard does not change while a player is thinking. The game only changes when a player makes a move.
- Finally, chess is a multi-agent environment because two players compete each other, and each player's decisions affect the opponent's strategy and the outcome of the game.

Solve the Water Jug problem with a 5-gallon jug and a 3-gallon jug. How can you measure exactly 4 gallons of water using only these two jugs and a pump? List your explicit assumptions and show the sequence of states from start to goal.

Ans)

ASSUMPTIONS:

- * Unlimited water supply.
- * Jugs have capacities of 5 gallons and 3 gallons.
- * Jugs can be completely filled, emptied or water can be poured from one jug to another until one is full or the other is empty.
- * Goal: Measure exactly 4 gallons.

STATE REPRESENTATION:

STEP	STATE	ACTION
1	(0,0)	Start
2	(5,0)	Fill 5-gallon Jug
3	(2,3)	Pour 5 $\rightarrow$ 3
4	(2,0)	Empty 3-gallon Jug
5	(0,2)	Pour remaining 2 into 3
6	(5,2)	Fill 5-gallon Jug
7	(4,3)	Pour 5 $\rightarrow$ 3 until 3 gallon is full.

Goal Achieved: (4,3)

The 5-gallon Jug contains exactly 4 gallons.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding, misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps, multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate, correct final answer	Minor calculation errors	Frequent errors, incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding, misses key elements	Misinterprets or unable to understand problem